TorchJD: Jacobian Descent in PyTorch

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GitHub
github.com/TorchJD/torchjd

TorchJD
torchjd.org

arXiv
arxiv.org/pdf/2406.16232

Motivation

Partial order

Define the partial order $(\mathbb{R}^m, \preceq)$, for $\boldsymbol{u}, \boldsymbol{v} \in \mathbb{R}^m$, as

$$\begin{aligned} & \boldsymbol{u} \preceq \boldsymbol{v} & \\ \Leftrightarrow \quad & u_i \leq v_i & \forall i \in [m] \end{aligned}$$

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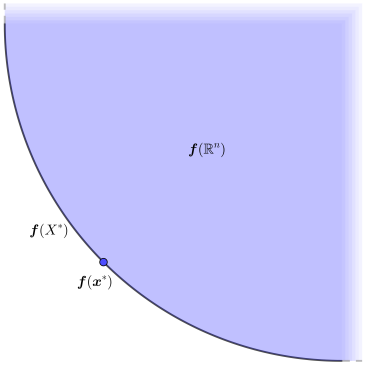
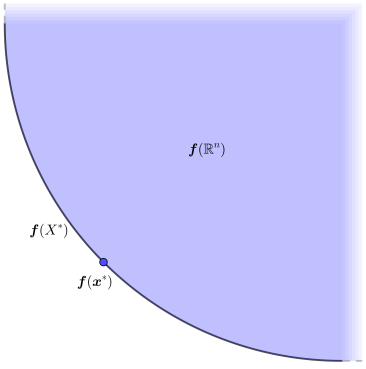
- $\begin{bmatrix} 2 \\ 1 \end{bmatrix} \preceq \begin{bmatrix} 2 \\ 2 \end{bmatrix}$
- $\begin{bmatrix} 2 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ are not comparable with this partial order.

Pareto optimality



Pareto optimality

Pareto set: X^*



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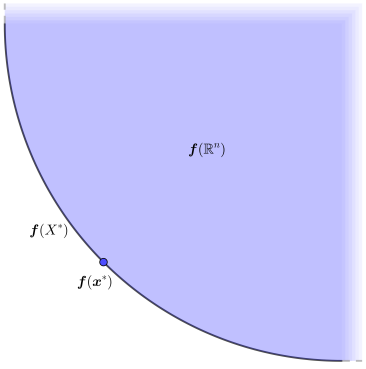

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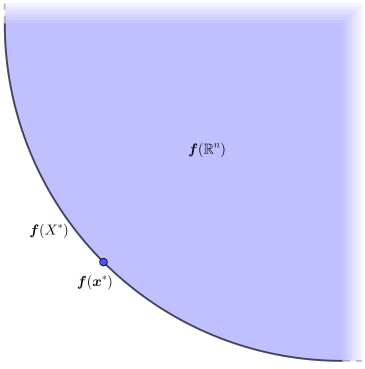

$$\boldsymbol{x}^* \in X^* \quad \Leftrightarrow \quad \forall \boldsymbol{y} \in \mathbb{R}^n,\ \boldsymbol{f}(\boldsymbol{y}) \preceq \boldsymbol{f}(\boldsymbol{x}^*) \rightarrow \boldsymbol{f}(\boldsymbol{y}) = \boldsymbol{f}(\boldsymbol{x}^*)$$

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We select $\boldsymbol{y} = -\eta \mathcal{A}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x})) \in \mathbb{R}^n$.

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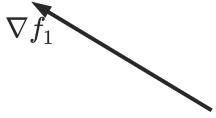
Algorithm 1: Jacobian descent with aggregator $\mathcal{A}$

Input: $\boldsymbol{x} \in \mathbb{R}^n$, $0 < \eta$, $T \in \mathbb{N}$, $\mathcal{A} : \mathbb{R}^{m \times n} \to \mathbb{R}^n$
for $t \leftarrow 1$ **to** T **do**
  $\boldsymbol{x} \leftarrow \boldsymbol{x} - \eta \mathcal{A}(\mathcal{J}\boldsymbol{f}(\boldsymbol{x}))$
Output: $\boldsymbol{x}$

Jacobian descent



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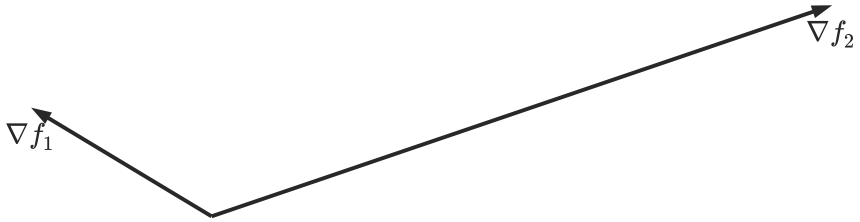

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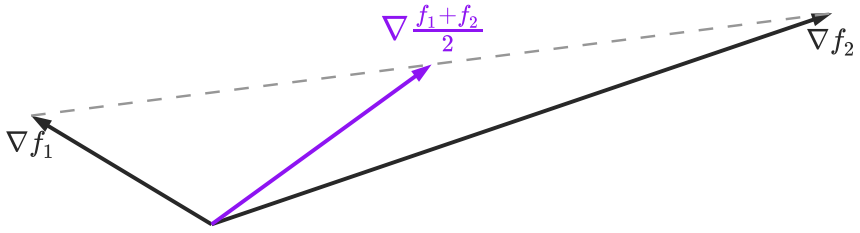

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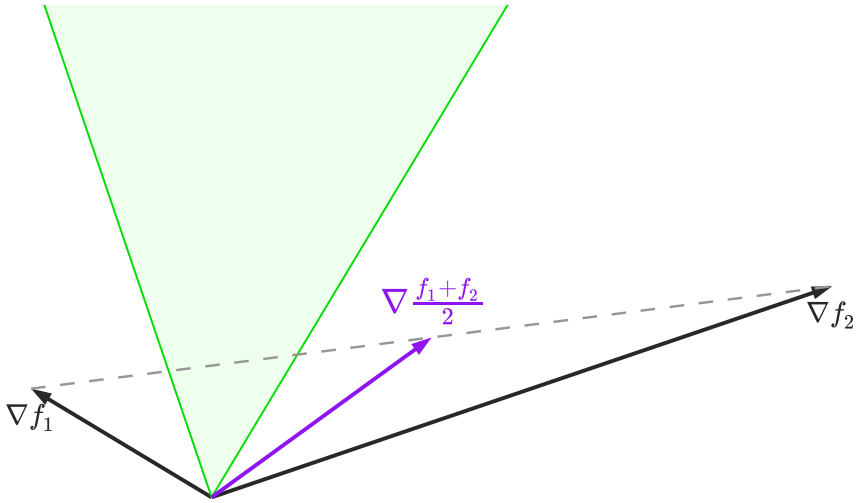

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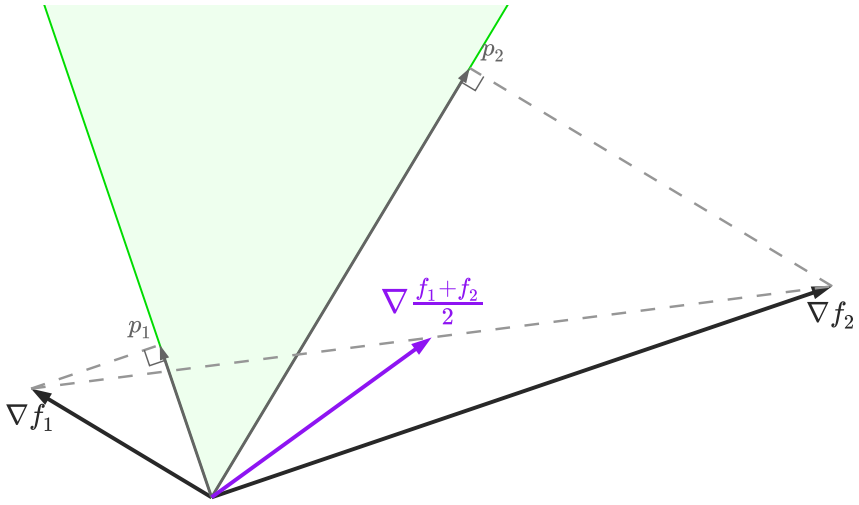

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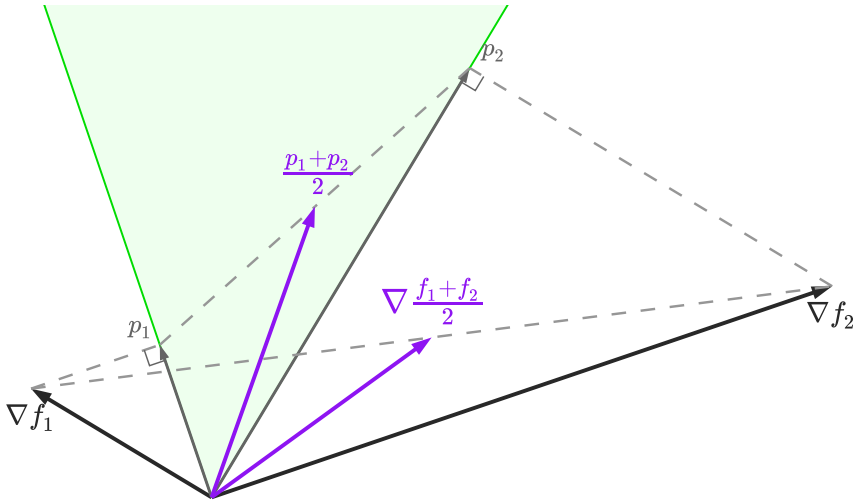
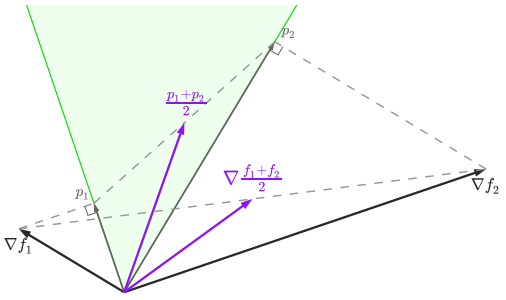


Formal definition of UPGrad

Given $J \in \mathbb{R}^{m \times n}$, let C_J be the cone $\{\boldsymbol{y} \in \mathbb{R}^n : \boldsymbol{0} \preceq J\boldsymbol{y}\}$.

$$p_i = \underset{\boldsymbol{y} \in C_J}{\arg\min} \|\nabla f_i - \boldsymbol{y}\|^2$$

$$\mathcal{A}_{\text{UPGrad}}(J) = \frac{1}{m} \sum_{i \in [m]} p_i$$



TorchJD

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`torchjd`	`autojac.jac_to_grad`	aggregate `.jac` fields into `.grad`

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  optimizer.step()
  optimizer.zero_grad()
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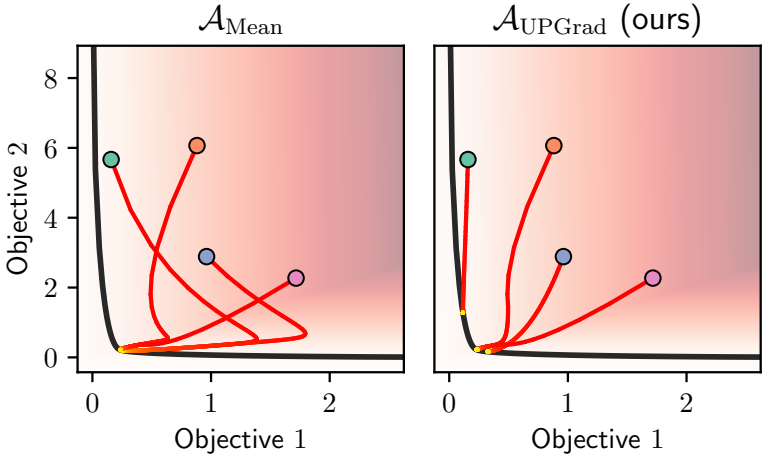

Discussion